

Schamchatt Application Architecture Document

This specification details the core client-side code structure, database entities, custom state management hooks, and UI components for the Schamchatt Platform.

1. Overview & System Design

The architecture is built around dynamic user engagement through real-time communication, gamified currency systems (Coins), community leaderboards, and extended user profiles.

Architectural Layers

- Presentation Layer:** React (18+) built with Tailwind CSS for utility-first responsive styling and Lucide Icons for dynamic visual cues.
- State & Hook Layer:** Custom React hooks (`useWallet`) providing localized state synchronization, optimism updates, and transactional safety abstractions.
- Data Model Layer:** Normalized entity models mapped for SQL/NoSQL persistence (e.g., PostgreSQL or Supabase / Firebase).

2. Database Schema & Entity Definitions

2.1 User Profile Entity (`UserProfile`)

Field Name	Data Type	Constraints	Description
<code>id</code>	<code>UUID</code>	Primary Key, Default ge...	Unique userIdentifier
<code>username</code>	<code>VARCHAR(32)</code>	Unique, Not Null	Display handle

Field Name	Data Type	Constraints	Description
avatar_url	TEXT ▾	Nullable ▾	Public URL to image asset
bio	TEXT ▾	Nullable ▾	Short user biography
level	INTEGER ▾	Default 1, Not Null ▾	Calculated level based on activity
created_at	TIMESTAMP ▾	Default NOW() ▾	Registration timestamp

2.2 User Wallet Entity (UserWallet)

Field Name	Data Type	Constraints	Description
wallet_id	UUID ▾	Primary Key, Default gen_random_uuid()	Unique wallet identifier
user_id	UUID ▾	Foreign Key (UserProfile.id), Unique	Associated profile owner
balance	BIGINT ▾	Default 0, Check (balance >= 0)	Current coin balance
updated_at	TIMESTAMP ▾	Default NOW()	Last balance update timestamp

2.3 Leaderboard Entry Entity (LeaderboardEntry)

Field Name	Data Type	Constraints	Description
entry_id	UUID ▾	Primary Key	Unique ranking record ID

Field Name	Data Type	Constraints	Description
user_id	UUID ▾	Foreign Key (UserProfile.id)	Target user
score	BIGINT ▾	Not Null, Default 0	Ranking metric (coins earned/spent)
rank_type	VARCHAR(16) ▾	Not Null ('weekly', 'all_time')	Leaderboard category scope
updated_at	TIMESTAMP ▾	Default NOW()	Last score modification

3. Wallet Hook Specification

useWallet.jsx

```
x
import { useState, useCallback } from 'react';

export const useWallet = (initialBalance = 2500) => {
  const [balance, setBalance] = useState(initialBalance);
  const [transactions, setTransactions] = useState([]);
  const [loading, setLoading] = useState(false);

  const addCoins = useCallback((amount, description = 'Purchase') => {
    setLoading(true);
    setTimeout(() => {
      setBalance((prev) => prev + amount);
      setTransactions((prev) => [
        {
          id: Date.now(),
          type: 'CREDIT',
          amount,
```

```

description,
timestamp: new Date().toISOString(),
},
...prev,
]);
setLoading(false);
}, 300);
}, []);

```

```

const deductCoins = useCallback((amount, description = 'Spent') ⇒ {
  if (balance < amount) return false;
  setBalance((prev) ⇒ prev - amount);
  setTransactions((prev) ⇒ [
    {
      id: Date.now(),
      type: 'DEBIT',
      amount,
      description,
      timestamp: new Date().toISOString(),
    },
    ...prev,
  ]);
  return true;
}, [balance]);

```

```

return { balance, transactions, loading, addCoins, deductCoins };
};

```

4. Leaderboard Component

```

#### `Leaderboard.jsx`

```

```

``jsx

```

```

import React, { useState } from 'react';

import { Trophy, Award, Crown, Flame } from 'lucide-react';

const MOCK_LEADERBOARD = [

  { id: '1', username: 'Alex_Master', coins: 45200, level: 42, avatar:
'https://images.unsplash.com/photo-1535713875002-d1d0cf377fde?w=100' },

  { id: '2', username: 'Sarah_Gamer', coins: 38900, level: 38, avatar:
'https://images.unsplash.com/photo-1494790108377-be9c29b29330?w=100' },

  { id: '3', username: 'PixelNinja', coins: 29400, level: 31, avatar:
'https://images.unsplash.com/photo-1570295999919-56ceb5ecca61?w=100' },

  { id: '4', username: 'CrypticUser', coins: 18200, level: 24, avatar:
'https://images.unsplash.com/photo-1507003211169-0a1dd7228f2d?w=100' },

];

export const Leaderboard = ({ onSelectUser }) => {

  const [filter, setFilter] = useState('weekly');

  return (

    <div className="w-full max-w-2xl bg-white dark:bg-slate-900 rounded-xl border
border-slate-200 dark:border-slate-800 shadow-sm overflow-hidden">

      <div className="p-6 border-b border-slate-200 dark:border-slate-800 flex
items-center justify-between">

        <div className="flex items-center space-x-3">

          <Trophy className="w-6 h-6 text-amber-500" />

          <h2 className="text-lg font-bold text-slate-900 dark:text-white">Community
Leaderboard</h2>

```

</div>

<div className="flex bg-slate-100 dark:bg-slate-800 rounded-lg p-1">

<button

onClick={() => setFilter('weekly')}

className={`px-3 py-1 text-xs font-semibold rounded-md transition-all \${

filter === 'weekly'

? 'bg-white dark:bg-slate-700 text-slate-900 dark:text-white shadow-sm'

: 'text-slate-500 hover:text-slate-900'

}`}

>

Weekly

</button>

<button

onClick={() => setFilter('all_time')}

className={`px-3 py-1 text-xs font-semibold rounded-md transition-all \${

filter === 'all_time'

? 'bg-white dark:bg-slate-700 text-slate-900 dark:text-white shadow-sm'

: 'text-slate-500 hover:text-slate-900'

}`}

>

All-Time

</button>

</div>

</div>

<div className="divide-y divide-slate-100 dark:divide-slate-800">

{MOCK_LEADERBOARD.map((user, index) ⇒ (

<div

key={user.id}

onClick={() ⇒ onSelectUser && onSelectUser(user)}

className="p-4 flex items-center justify-between hover:bg-slate-50
dark:hover:bg-slate-800/50 cursor-pointer transition-colors"

>

<div className="flex items-center space-x-4">

<div className="w-8 flex justify-center font-bold text-slate-400">

{index === 0 && <Crown className="w-5 h-5 text-amber-500" />}

{index === 1 && <Award className="w-5 h-5 text-slate-400" />}

{index === 2 && <Award className="w-5 h-5 text-amber-700" />}

{index > 2 && index + 1}

</div>

```
    <img src={user.avatar} alt={user.username} className="w-10 h-10 rounded-full
object-cover" />
```

```
    <div>
```

```
        <p className="font-semibold text-slate-900 dark:text-white
text-sm">{user.username}</p>
```

```
        <span className="inline-flex items-center px-2 py-0.5 rounded text-xs
font-medium bg-slate-100 text-slate-800 dark:bg-slate-800 dark:text-slate-300">
```

```
            Lvl {user.level}
```

```
        </span>
```

```
    </div>
```

```
</div>
```

```
<div className="flex items-center space-x-1 text-amber-500 font-bold text-sm">
```

```
    <Flame className="w-4 h-4 fill-amber-500" />
```

```
    <span>{user.coins.toLocaleString()}</span>
```

```
</div>
```

```
</div>
```

```
    )}
```

```
</div>
```

```
</div>
```

```
);
```

```
};
```

5. Profile & Coin Store Modals

5.1 User Profile Modal (**UserProfileModal.jsx**)

```
import React from 'react';

import { X, Coins, ShieldCheck, Calendar } from 'lucide-react';

export const UserProfileModal = ({ user, isOpen, onClose }) => {

  if (!isOpen || !user) return null;

  return (

    <div className="fixed inset-0 z-50 flex items-center justify-center bg-black/50
backdrop-blur-sm">

      <div className="w-full max-w-md bg-white dark:bg-slate-900 rounded-2xl
shadow-2xl overflow-hidden border border-slate-200 dark:border-slate-800">

        <div className="h-24 bg-gradient-to-r from-indigo-500 to-purple-600 relative">

          <button

            onClick={onClose}

            className="absolute top-3 right-3 p-1.5 rounded-full bg-black/20
hover:bg-black/40 text-white transition-colors"

            >

              <X className="w-4 h-4" />

            </button>
```

</div>

<div className="px-6 pb-6 relative">

<div className="-mt-12 mb-4 flex justify-between items-end">

<img

src={user.avatar || 'https://via.placeholder.com/100'}

alt={user.username}

className="w-20 h-20 rounded-full border-4 border-white
dark:border-slate-900 object-cover shadow-md"

/>

<span className="inline-flex items-center px-3 py-1 rounded-full text-xs
font-semibold bg-indigo-100 text-indigo-800 dark:bg-indigo-950 dark:text-indigo-300">

Level {user.level || 1}

</div>

<h3 className="text-xl font-bold text-slate-900 dark:text-white flex items-center
gap-1.5">

{user.username}

<ShieldCheck className="w-5 h-5 text-blue-500" />

</h3>

<p className="text-sm text-slate-500 dark:text-slate-400 mt-1">

```
{user.bio || 'Community member active in Schamchatt discussions.'}
```

```
</p>
```

```
<div className="mt-6 grid grid-cols-2 gap-4 border-t border-slate-100  
dark:border-slate-800 pt-4">
```

```
<div className="flex items-center space-x-3">
```

```
<Coins className="w-5 h-5 text-amber-500" />
```

```
<div>
```

```
<p className="text-xs text-slate-400">Total Balance</p>
```

```
<p className="text-sm font-bold text-slate-800 dark:text-slate-200">
```

```
{user.coins ? user.coins.toLocaleString() : 0}
```

```
</p>
```

```
</div>
```

```
</div>
```

```
<div className="flex items-center space-x-3">
```

```
<Calendar className="w-5 h-5 text-slate-400" />
```

```
<div>
```

```
<p className="text-xs text-slate-400">Joined</p>
```

```
<p className="text-sm font-bold text-slate-800  
dark:text-slate-200">Recent</p>
```

```
</div>
```

```

        </div>

    </div>

</div>

</div>

</div>

);

};

```

5.2 Coin Store Modal (**CoinStoreModal.jsx**)

```

import React from 'react';

import { X, Coins, Sparkles, ShoppingBag } from 'lucide-react';

const COIN_PACKS = [

  { id: 'pack_1', amount: 500, price: '$0.99', popular: false },

  { id: 'pack_2', amount: 2500, price: '$4.99', popular: true },

  { id: 'pack_3', amount: 10000, price: '$18.99', popular: false },

];

export const CoinStoreModal = ({ isOpen, onClose, onPurchase }) => {

  if (!isOpen) return null;

  return (

    <div className="fixed inset-0 z-50 flex items-center justify-center bg-black/50
backdrop-blur-sm">

```

```
<div className="w-full max-w-lg bg-white dark:bg-slate-900 rounded-2xl
shadow-2xl overflow-hidden border border-slate-200 dark:border-slate-800">
```

```
<div className="p-6 border-b border-slate-100 dark:border-slate-800 flex
justify-between items-center">
```

```
<div className="flex items-center space-x-2">
```

```
<ShoppingBag className="w-5 h-5 text-amber-500" />
```

```
<h2 className="text-lg font-bold text-slate-900 dark:text-white">Schamchatt
Store</h2>
```

```
</div>
```

```
<button onClick={onClose} className="text-slate-400 hover:text-slate-600">
```

```
<X className="w-5 h-5" />
```

```
</button>
```

```
</div>
```

```
<div className="p-6 grid grid-cols-3 gap-4">
```

```
{COIN_PACKS.map((pack) => (
```

```
<div
```

```
key={pack.id}
```

```
className={`relative rounded-xl p-4 flex flex-col items-center justify-between
border text-center transition-all hover:scale-105 ${
```

```
pack.popular
```

```
? 'border-indigo-500 bg-indigo-50/50 dark:bg-indigo-950/20'
```

```

      : 'border-slate-200 dark:border-slate-800 bg-white dark:bg-slate-900'
    }}

  >

  {pack.popular && (

    <span className="absolute -top-3 px-2 py-0.5 bg-indigo-600 text-white
text-[10px] font-bold rounded-full flex items-center gap-1">

      <Sparkles className="w-3 h-3" /> Popular

    </span>

  )}

  <Coins className="w-8 h-8 text-amber-500 my-2" />

  <div>

    <p className="font-extrabold text-slate-900 dark:text-white text-base">

      {pack.amount.toLocaleString()}

    </p>

    <p className="text-xs text-slate-400">Coins</p>

  </div>

  <button

    onClick={() => onPurchase(pack)}

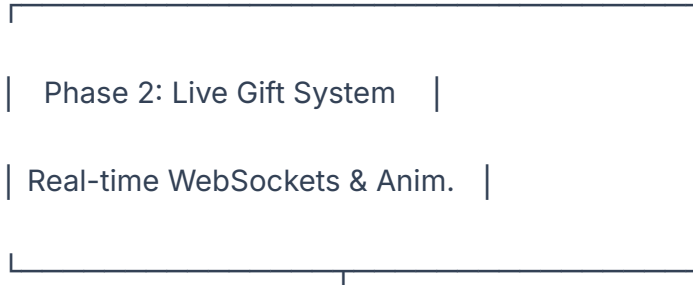
```

Next Implementation Phases

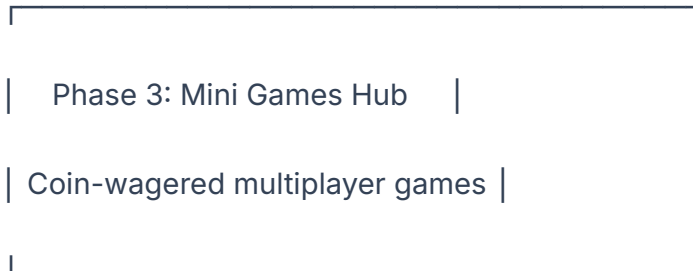
Phase 1: Core Modules

Wallet, Profile & Leaderboard

|



|



1. **State Integration:** Connect `useWallet.jsx` hooks to global context managers (e.g., Zustand or Redux Toolkit) to synchronize wallet state across messaging views.
2. **Gift System Architecture (Phase 2):** Introduce dynamic streaming components where users can send animated gifts funded directly via the `deductCoins` API.
3. **Mini Games Infrastructure (Phase 3):** Implement multiplayer WebSocket-backed session rooms for turn-based coin games.